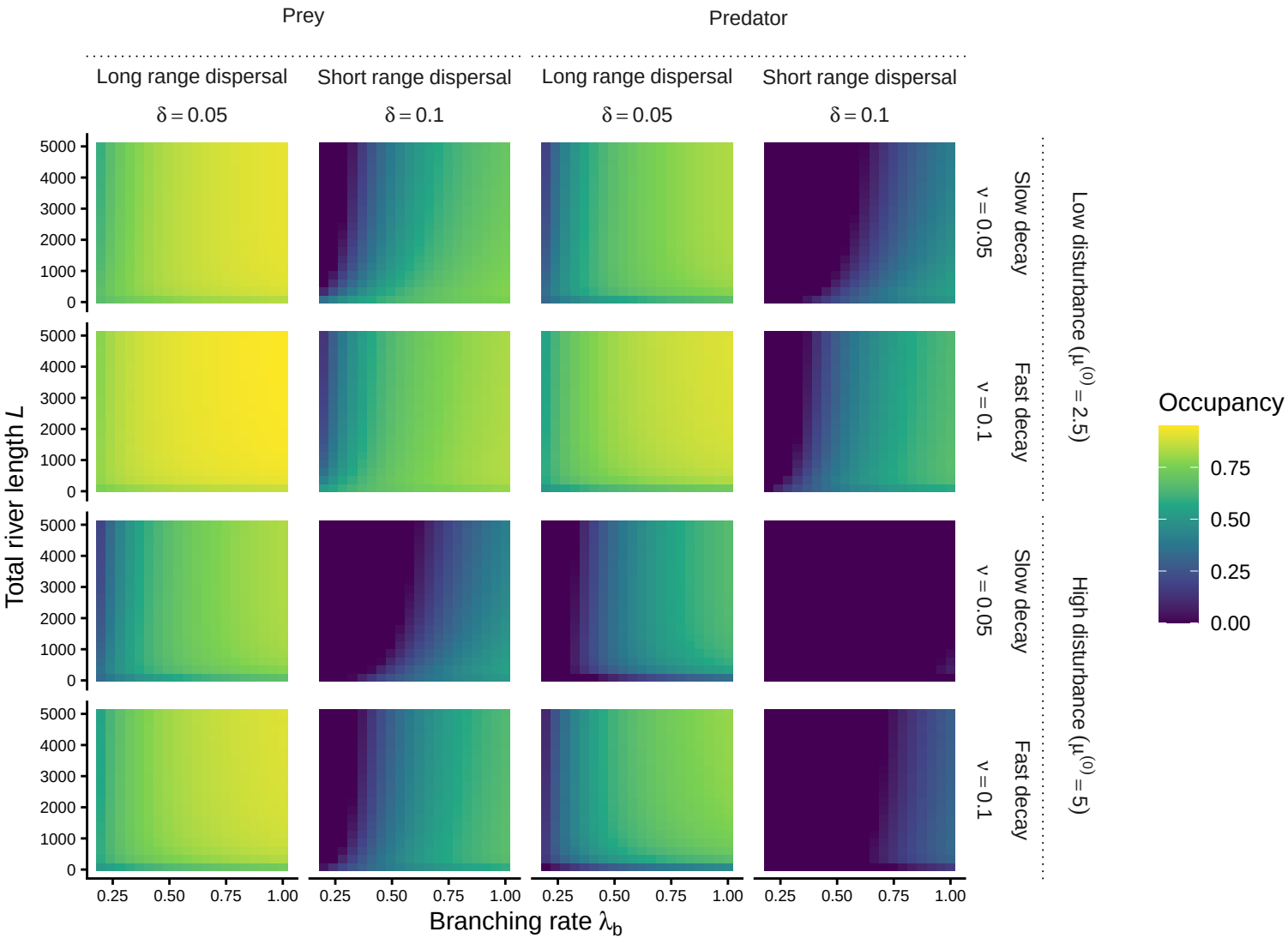
Prey
Predator
Long range dispersal
Short range dispersal
Long range dispersal
Short range dispersal
δ = 0.05
δ = 0.1
δ = 0.05
δ = 0.1
Slow decay
ν = 0.05
Fast decay
ν = 0.1
Slow decay
ν = 0.05
Fast decay
ν = 0.1
Low disturbance (μ(0) = 2.5)
High disturbance (μ(0) = 5)
Total river length L
0
1000
2000
3000
4000
5000
0.25
0.50
0.75
1.00
Branching rate λb
Occupancy
0.00
0.25
0.50
0.75